

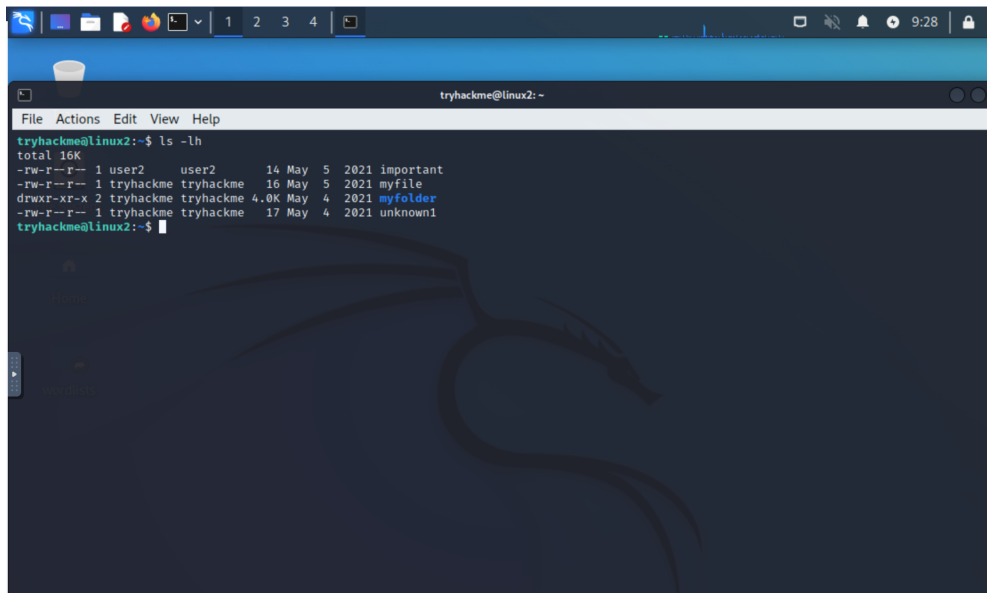
FILE SYSTEM PERMISSIONS

Certain users cannot access certain files or folders.

Some commands that can be used to determine what access we have and where it leads us.

Examples:

Ls -lh- list contents of the current directory when using the -l switch.



```
tryhackme@linux2:~$ ls -lh
total 16K
-rw-r--r-- 1 user2  user2   14 May  5  2021 important
-rw-r--r-- 1 tryhackme tryhackme 16 May  5  2021 myfile
drwxr-xr-x 2 tryhackme tryhackme 4.0K May  4  2021 myfolder
-rw-r--r-- 1 tryhackme tryhackme 17 May  4  2021 unknown1
tryhackme@linux2:~$
```

A file or folder can have characteristic that determine both what action are allowed and what user or group has that ability to perform the given action.

These actions are:

- Read
- Write
- Execute

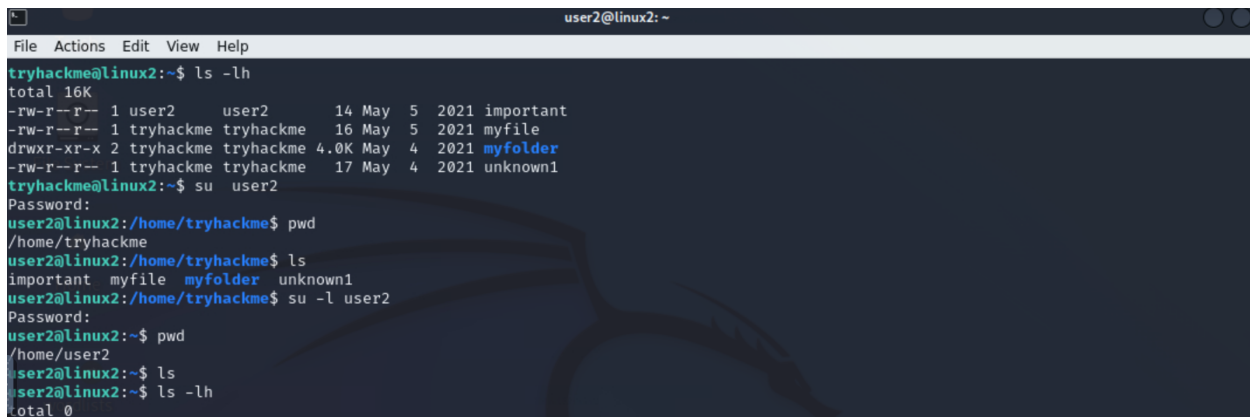
Switching Between users

SU ~command used unless you are the root user, the you are required to know two things to facilitate this transition of user account:

- The user we wish to switch to
- The user's password

Su command-execute a command once you log in or specify a specific shell to use.

Command ~su user2

A terminal window titled 'user2@linux2: ~' showing a sequence of commands and outputs. The user 'tryhackme' runs 'ls -lh' showing file permissions for 'important', 'myfile', 'myfolder', and 'unknown1'. Then they run 'su user2', enter a password, and run 'pwd' showing '/home/tryhackme'. Next, they run 'ls' showing the same files. Then they run 'su -l user2', enter a password, and run 'pwd' showing '/home/user2'. Finally, they run 'ls' and 'ls -lh', both showing an empty directory with 'total 0'.

```
user2@linux2: ~  
File Actions Edit View Help  
tryhackme@linux2:~$ ls -lh  
total 16K  
-rw-r--r-- 1 user2      user2      14 May  5 2021 important  
-rw-r--r-- 1 tryhackme tryhackme  16 May  5 2021 myfile  
drwxr-xr-x 2 tryhackme tryhackme  4.0K May  4 2021 myfolder  
-rw-r--r-- 1 tryhackme tryhackme  17 May  4 2021 unknown1  
tryhackme@linux2:~$ su user2  
Password:  
user2@linux2:/home/tryhackme$ pwd  
/home/tryhackme  
user2@linux2:/home/tryhackme$ ls  
important myfile myfolder unknown1  
user2@linux2:/home/tryhackme$ su -l user2  
Password:  
user2@linux2:~$ pwd  
/home/user2  
user2@linux2:~$ ls  
user2@linux2:~$ ls -lh  
total 0
```

FILE PERMISSIONS IN NUMERIC FORMAT

Every file and directory has a set of permissions that control who can read, write and execute it.

Each letter represents a specific permission:

- R =read
- W =write
- X=execute

Converting symbolic permissions to numbers

All permissions appears in a series of 3 – rwx

- 0- No permissions (~) {---}
- 1- Execute (x) {--x}
- 2- Write(w) {-w-}
- 3- 2+1 (write+Execute) {-wx}
- 4- Read (r) {r--}
- 5- 4+1 (Read+Execute) {rx-}
- 6- 4+2 (read +write) {rw}
- 7- 4+3 (read+write+execute) {rwx}

Target ID	USER			GROUP			EXTERNAL USER			
0	~	~	~	~	~	~	~	~	~	No permissions (-)
1	~	~	x	~	~	x	~	~	x	Execute (~-x)
2	~	w	~	~	w	~	~	w	~	Write (~w-)
3	~	w	x	~	w	x	~	w	x	Write ,execute (~wx)
4	r	~	~	r	~	~	r	~	~	Read (r~-)
5	r	~	x	r	~	x	r		x	Read and execute(rx~-)
6	r	w	~	r	w	~	r	w	~	Read write(rw~-)
7	r	w	x	r	w	x	r	w	x	rwX

Example:

664~(rw~ ww~ r~-)

775~(rwx rwx r~x)

356~(-wx r~x rw~)

777~(wXw rwx rwx)